

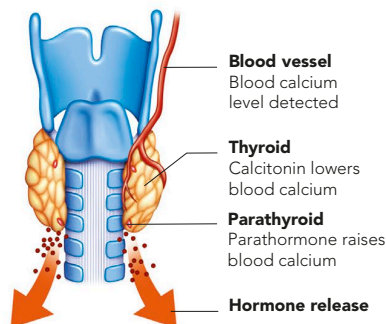
HORMONAL ACTION

HORMONES REGULATE THE FUNCTION OF THEIR TARGET CELLS BY ADJUSTING THE RATE AT WHICH A CELL'S BIOCHEMICAL REACTIONS OCCUR. DIFFERENT HORMONES ARE RELEASED ACCORDING TO DIFFERENT TRIGGER MECHANISMS.

HORMONAL TRIGGERS

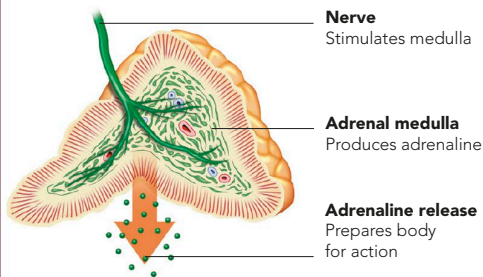
Various stimuli cause an endocrine gland to release more of its hormone. In some cases, the gland responds to the level of a certain substance in the blood, using a feedback loop (see opposite). In other cases, there is an intermediate mechanism, such as the hypothalamus–pituitary complex. The adrenal gland is controlled both by adrenocorticotropic

hormone (ACTH), released by the pituitary on cue from the hypothalamus, and by nerve impulses direct from the hypothalamus. The pea-sized pineal gland, near the centre of the brain, is triggered by darkness to release the sleep hormone melatonin. Pineal activity is inhibited by light, which is detected by the eye and sent to the gland as a series of nerve impulses.



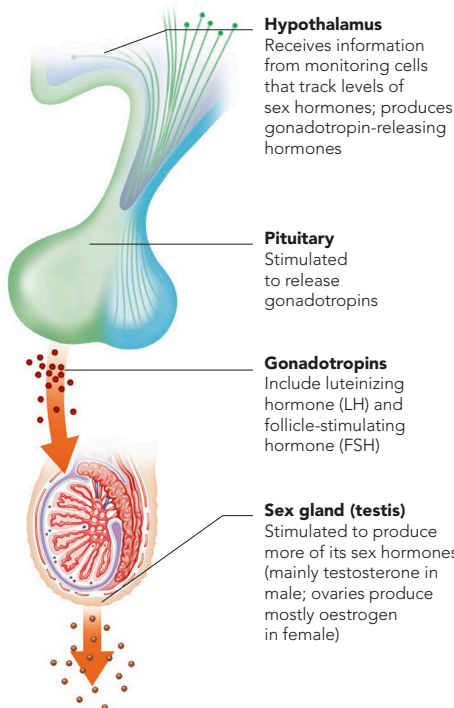
BLOOD LEVEL STIMULATION

Low blood calcium levels inhibit release of calcitonin from the thyroid and stimulate the parathyroids to release parathormone; calcium levels are raised.



DIRECT INNERVATION

The adrenal medulla receives nerve fibres (is innervated) from the hypothalamus via the sympathetic nervous system.



HYPOTHALAMIC–PITUITARY CONTROL

As sex hormone levels fall, gonadotropin-releasing hormones (GnRH) are sent from the hypothalamus to the pituitary, which releases more gonadotropins.